

# EasyA Mathematics Team Mathematics Challenge

Presented by EasyA  Organized by A lot of people

Last Updated: January 14, 2022

EasyA first mathematics competition is here! It will be held online at **$15^{th}$ January, 2022, 3:00-3:45 pm.**

The following document outlines some information competition.

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# 1 About the competition

## 1.1 Aim

The EasyA Mathematics Team building competition is inspired by Team Mathematics Challenge, hosted by the UK Mathematics Trust. We want to give different students the opportunity to tackle math questions and have a taste of what university mathematics is like while developing teamwork and communication skills.

We also aim to help secondary school students to form connections with like-minded people who are interested in problem solving.

## 1.2 Disclaimer

For the latest updates in our competition, please join our Discord server: https://discord.gg/jUn8bahCJx

By submitting this form you agree to our privacy policy: http://easya.co.uk/privacy/

## 1.3 Contacts and Organizers

- Phil Kwok, Founder of EasyA,
  Email: phil@easya.co.uk.
  LinkedIn: https://www.linkedin.com/in/philip-k-building/
- Ryan Lam (Lead Marker of the Competition), Year 2, University of Bristol,
  Email: ryan.cklam.2020@bristol.ac.uk
  LinkedIn: https://www.linkedin.com/in/ryan-lam-hk/
- Karanveer Singh, Moderator at the EasyA Discord Server
  LinkedIn: https://www.linkedin.com/in/karanveer-singh-kvs/
- Krish Lakhani, Head Moderator at the EasyA Discord Server
  LinkedIn: https://www.linkedin.com/in/krish-lakhani-09aa2316a/

# 2 Participating in the competition

## 2.1 Eligibility

The EasyA Team Maths Challenge is designed for all students studying mathematics from secondary school all around the world. From year 7 in GCSE, to year 13 in A Level. The following subsection will lay out measures that ensure everyone has a fair playing ground.

## 2.2 Entering the Competition

To enter, you need to form a team of 2 to 4 people. After that, **all team members** need to fill in the google form: https://forms.gle/3ZynQFmFWTj4fJ3u6 to register your interest.

## 2.3 Format of Competition

Teams are given a paper that contains 20 questions with varied difficulty:

- Question 1-10; Part A: GCSE Level Questions, 5 marks each.
- Question 11-16; Part B: Harder GCSE / AS Level Mathematics Questions, 10 Marks Each.
- Question 17-20; Part C: Difficult AS / A Level Mathematics Questions, 15 Marks Each.

**IMPORTANT: YEAR 13 STUDENTS ARE NOT ALLOWED TO ATTEMPT PART A QUESTIONS ALONE.**

And their aim is to divide up among themselves to answer the questions and get as many points as possible under 45 minutes (3:00pm - 3:45pm).
Remark:

- To earn full marks, you must show your work on all of your answers.
- Unless otherwise stated, all answers should be given EXACTLY.
- Each team will be given one answer booklet as a reference. *They can choose to not use it if they think there are better alternative to record their answers.*
- **Each team have to make ONE AND EXACTLY ONE submission of the paper.** In the answer paper, you must indicate the question that you have chosen and **who had answered the question**.

## 2.4 Submitting the Answer Paper

Please submit your paper as **one single PDF file** and name it according to your team's name. You can submit at your **Team's Chat Channel** under the competition tab at discord. If there are any technical difficulties, feel free to email us at the address above.

# 3 Topics Included

## Part A: GCSE Questions

Logical Deduction. Pattern Recognition. Basic Summary of statistics. Basic Algebraic Manipulation. Equations of Circle. Volume and Area. Pythagoras Theorem.

### Part B: Easier AS Level / A Level Questions

Algebraic Manipulation. Theorem and Properties of Circles. 3D Trigonometry. Complex Numbers and their Basic Properties. Combinatorics and Probability.

### Part C: Harder A Level Questions

Sequence and Series. Basic Calculus. Integration by Parts. Matrices. Mathematical Induction.

## 4 Marking and Release of Results

After submission of the paper, markers will began to mark each question your team had attempted.

- The marking will be returned to your team after the winners are announced. You can see your team's total marks, detailed marking for each question, and some additional advice and remarks when writing up your answers.
- Alongside with each team's marked script, a suggested marking script will be posted, outlining where are the points awarded.
- Points are awarded if your working shows clear and concise logical progression, even if the answer is wrong, partial credit might be awarded.
- If there have been multiple attempts of the same question, marks will be given to the most complete and correct attempt.
- Ryan Lam is the lead Marker of the Competition. Please forward your enquiries to ryan.cklam.2020@bristol.ac.uk if you have any questions regarding the marking.

## 5 Frequently Asked Questions

### What are the Prizes of this competition?

The winner of the competition will get a 40 pounds Amazon Voucher, divided across the teammates!

### How do I upload the Paper?

Send a message in your team's discord text channel! Remember to attach **one single PDF file** to your script. If you have any problem regarding submission, you can send an email to the organizers (refer to section 1.3).

# 6 Acknowledgements

Huge thanks to Phil for overseeing the discord server, Karanveer and Krish for contacting the teams in the server, and all the volunteer markers providing suggestions and pointing out typos.